THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL

FORM FOUR PRE-MOCK EXAMINATION
033 BIOLOGY 1
MARKING SCHEME 2025

SECTION A

Question 1

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	A	D	E	B	B	C	D	E	C

@01=10 marks

Question 2

LIST A	i	ii	iii	iv	v	vi
LIST B	E	G	F	B	A	D

@01=06 marks

SECTION B

Question 3

Classes of phylum Chordata:

i) class mammals, they have mammary gland, fur, and are warm blooded example rat
ii) class reptile. Their poikilothermic, have scale. Lay eggs, example lizard
iii) class amphibia, lay eggs in water, poikilothermic, no scales. example frog
iv) class aves, have feather, lay eggs, have beaks; example duck
v) class Osteichthyes, have scales, gills… example tilapia fish
vi)class Chondrichthyes, the skeleton is made of cartilage, the body is covered with placoid scales, the caudal fin (tail) has two lobes that differ is size; example sharks

$01^{1}/_{2}$ marks @ total = (09 marks)

Question 4

According to physical state, wastes can be classified as Solid wastes Liquid wastes Gaseous wastes and Sludge wastes.

Solid wastes

These are wastes which are about 75% solid. Example of solid wastes include house hold solid wastes such as vegetable, fruit and garden wastes,…

Liquid wastes

These are wastes which are in the form of liquid or watery materials. They contain less than 25% solid substances. Example of liquid wastes include water from sinks, wash basins and baths, and urine.

Gaseous wastes

These are waste materials which are released into the atmosphere in gaseous state. Examples of gaseous wastes are carbon dioxide, carbon monoxide…

Sludge wastes

These are semi-solid wastes i.e. waste which is neither solids nor liquids, for example, a mixture of livestock urine and cow dung

Any three, 01½ @ = 04 ½

Basic Principle of Waste Disposal (Principles of Waste Management)

The basic principles of waste management are reduction, re-use and recycling.

Reduction

Reduction involves uses of alternatives in order to prevent or reduce waste generation; for example, using a basket for shopping instead of using disposable plastic bags.

Re-use

Re-use involves the use of waste (used material) for other useful activities; for example, using a margarine container for storing sugar instead of throwing it away.

Recycling

Any three, 01½ @ = 04 ½ marks

Recycling involves the conversion of waste in other useful materials. Recycling include the sending back of plastic bottles to the factory for manufacturing of new plastic bottles, sending back of broken glass to the factory for manufacturing of new glass…

Question 5

- Supply of oxygen to tissues
- Supply of nutrients such as glucose, amino acids, and fatty acids
- Removal of waste such as carbon dioxide, urea, and lactic acid
- Body defense through circulation of white blood cells,
- Coagulation to stop bleeding.
- Transport of hormones
- Regulation of body pH
- Regulation of body temperature
- Hydraulic functions such as causing engorgement in specialized tissues resulting in an erection of that tissue; examples are the erectile tissue in the penis and clitoris.

Any six: 01$^{1}/_{2}$ marks @ total= (09 marks)

Question 6

Six factors affecting growth in plants are

(i) Mineral nutrients
(ii) Oxygen in the atmosphere used for respiration process
(iii) Carbon dioxide in the atmosphere important for photosynthesis process
(iv) Water
(v) Light
(vi) Hormones

01$^{1}/_{2}$ marks @ total= (09 marks)

Question 7

(a) Plants do not have complex excretory organs like those of animals due to the following reasons:

- Plants synthesize (manufacture) most their materials with respect to the plants' needs, therefore less excess material to be considered as wastes.
- Plants re-use most of the waste products; for example, oxygen and carbon dioxide are both used for respiration and photosynthesis respectively.
- Plants metabolic rate is low as they are not as active as animals are, therefore producing less metabolic wastes.

@1½ marks = 04½ marks

(b)

@0½ marks = 04½ marks

Excretory Products	Metabolism	Excretory structure	Example of Organism
Carbon dioxide and water	**Respiration**	**Lungs, skin**	Man, frog, bird
		Stomata	Plants
Oxygen	Photosynthesis	**Stomata**	Plants
Ammonia	Protein metabolism	**Cell membrane**	Amoeba
Urea		**Kidney, skin**	Man
Uric acid		**Kidney**	Lizard, bird
Bile pigments	Haemoglobin breakdown	**Liver**	Man

Question 8

(a) Complete dominance Is the dominance where by one gene mask the expression of the other gene.
A dominant gene always masks a recessive gene when the two occur together
Incomplete dominance: there is no dominant or recessive gene, but both express themselves partially. It results in a heterozygous individual which does not resemble any of the heterozygous individual which does not resemble any
03 marks
Co-dominance: In co- dominance genes from both parents are dominant and are phenotypically expressed in the offspring.

(b) Solution

i. Let, R be Red and W be White;

RR (Red) X WW (White)

	R	R
W	RW	RW
W	RW	RW

02 marks

Where by: RW = Pink

F1 = RW, RW, RW, RW

Genotype: All RW *00½ marks*

ii.

RW X RW

	R	W
R	RR	RW
W	RW	WW

02 marks

Where by: RR =Red, RW = Pink and WW White

F1 = RR, RW, RW, WW

Genotype: 1:2:1 *00½ marks*

Incomplete dominance *01 marks*

SECTION C

Question 9

Intro: definition of mouth as part of alimentary canal/digestive system *01½ marks*

How is the human mouth adapted for digestion of food:

- Ha teeth for chewing the food (The teeth are very hard objects and are well differentiated for their function. There are incisors for cutting off food (soft food), canines for tearing flesh (poorly developed in humans) and pre-molars and molars for crushing the food.)
- There is saliva which has the enzyme ptyalin. Ptyalin is a salivary amylase which digests starch to produce maltose. The saliva also provides the optimum medium for the enzyme ptyalin to function, and it also moisten and lubricates the food and making it easy to swallow.
- There is also the tongue which helps in mixing up the food and shapes it into bolus,
- epiglottis which shuts off the air way (trachea) to prevent food from entering it during swallowing.

Dental caries *@01½ marks = 06 marks*

This occurs when bacteria destroy the outer part of the tooth due to the formation of plaque and consequently acids. *02 marks*

Signs and symptoms include tooth ache, holes in the teeth and bad breath.

Control is by brushing one's teeth, using mouth wash; minimize the eating of too sugary foods, having regular dental checkup and by removing of affected tooth. *02 marks* *02 marks*

Conclusion: Mouth (Buccal cavity) is very important as part of digestion system therefore it should be well taken care of to avoid dental caries, which can… *01½ marks*

Question 10

Drug abuse is the indiscriminate and excessive use of psychoactive drugs for non-medical reasons with no regard to their side effects. Dependence caused by regular use of psychoactive drugs leads to drug addiction.

01 mark

Effects of Drug Abuse

Health Hazards:

- Cigarette smoking can lead to lung cancer and heart diseases.
- Alcohol can cause brain damage.
- Some drugs cause ulcers and rotten teeth.
- Many drugs damage the brain and other organs.
- Many drugs lead to addiction.
- Some drugs may lead to kidney damage.
- Pregnant women can affect their unborn baby.

Socio-Economic Hazards of Drug Abuse:

Any six @01½ marks

- Drug addicts fail to concentrate at work (poor outcome).
- Drugs weaken our bodies- poor development of the society.
- Addicts indulge in irresponsible behaviour which can lead to unwanted pregnancies, STD's and HIV/AIDS.
- Student addicts mostly drop out of school or absent themselves.
- May lead to traffic offences and accidents.
- Economic burden to others as the addicts do not work.
- Drug addiction can lead to marriage breakdown.
- Drug addiction lead to more crime.

Measures to Control Drug Abuse

- Avoid taking any drug without doctor's prescription.
- Keep off from peer pressure and addicted people.
- Be principled and say no to anyone who offers the drug to you.
- Use your leisure time properly (for sports and hobbies).
- Inform community leader or police of any use or trafficking of drugs.
- Form counselling club to educate people on how to keep off from drugs (education on proper use of drugs)

Any four @01 mark

- Addicts should seek help from health officials.

Conclusion (Any relevant) *01 mark*

Question 11

Importance of providing care and support to people living with HIV/AIDS(PLWHA) are

- Gives hope to the victim
- Reduce stress of the victim
- Increases life expectancy of the victim
- Helps victim to engage in economic activities

Introduction =01$^{1}/_{2}$ mark
Main body =03
marks @ total= (12 marks)
Conclusion=01$^{1}/_{2}$ marks